Is the Shark a Wolf to Man?

Analysis of the `shark_attacks.csv` dataset

First Name Last Name – L1 SEA

1 Data and method

The `shark_attacks.csv` file lists documented shark attacks, each with its date, type (*Provoked*, *Unprovoked*, *Boat/Boating*, *Sea Disaster*, *Invalid*), country, region (*area*), exact location (*location*), the victim's activity and the time. The whole analysis relies on **pivot tables** (using "count" rather than "sum"), filtered on years from 1900 onwards, combined with conditional formatting (colour scales) to highlight extreme values. The month was extracted from the date with `=MONTH(B2)`; times that were too vague ("morning", "afternoon") and rare activities (fewer than 10 occurrences) were discarded.

2 Evolution of attacks since 1900

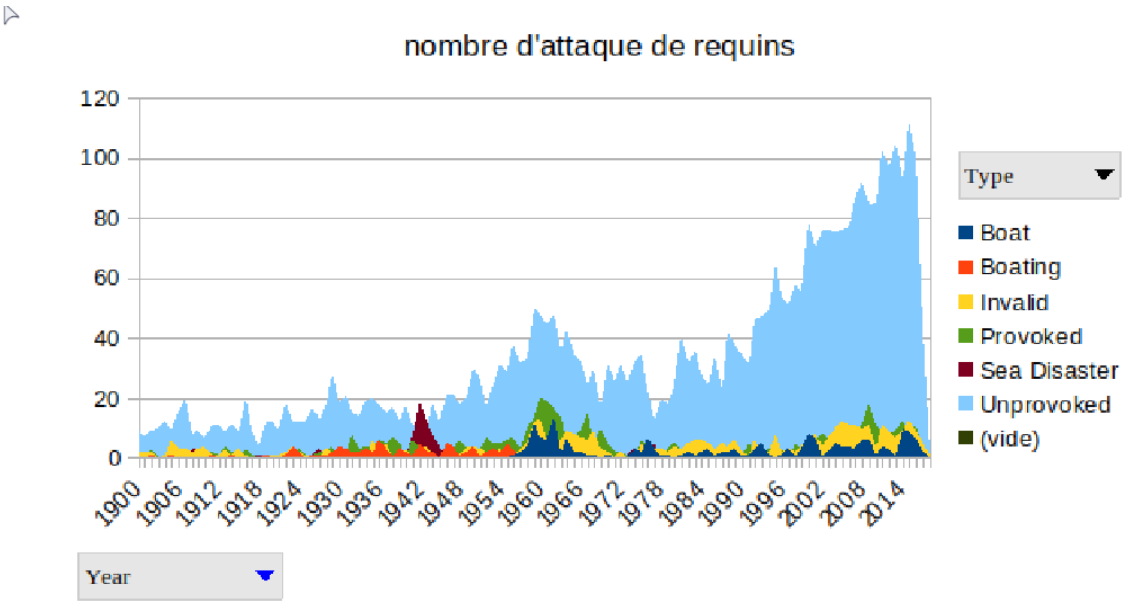


Figure 1: Number of shark attacks per year and by category (1900–2016).

Unprovoked attacks clearly dominate and rise sharply over the period: about twenty per year at the beginning of the century, more than 100 per year in the 2010s. A first peak appears in the 1950s–1970s, followed by an almost continuous increase from 1990 onwards. A polynomial trend line (degree 15) fits the data with $R^2 = 0.929$. The other categories remain stable, except for a peak of *Sea Disasters* around 1942, which corresponds to shipwrecks during the Second World War. This increase is most likely explained by the spread of holidays and water sports: more people in the water means more encounters.

3 Provoked or unprovoked attacks?

Over the period, there are **3,951 unprovoked attacks** against **531 provoked ones**. The provoked/unprovoked ratio is $531/3951 \approx 13.4\,\%$; as a share of the total of both categories, provoked attacks only account for about $11.8\,\%$. In the vast majority of cases, the victim therefore did nothing to trigger the attack.

4 Where do sharks attack?

Down at beach level, one area stands out very clearly: **New Smyrna Beach (Volusia County, Florida)** with **162 attacks**, followed by Daytona Beach (29) and Ponce Inlet (18), all three neighbouring each other in the same county. Elsewhere, individual locations usually total only between 1 and 5 attacks.

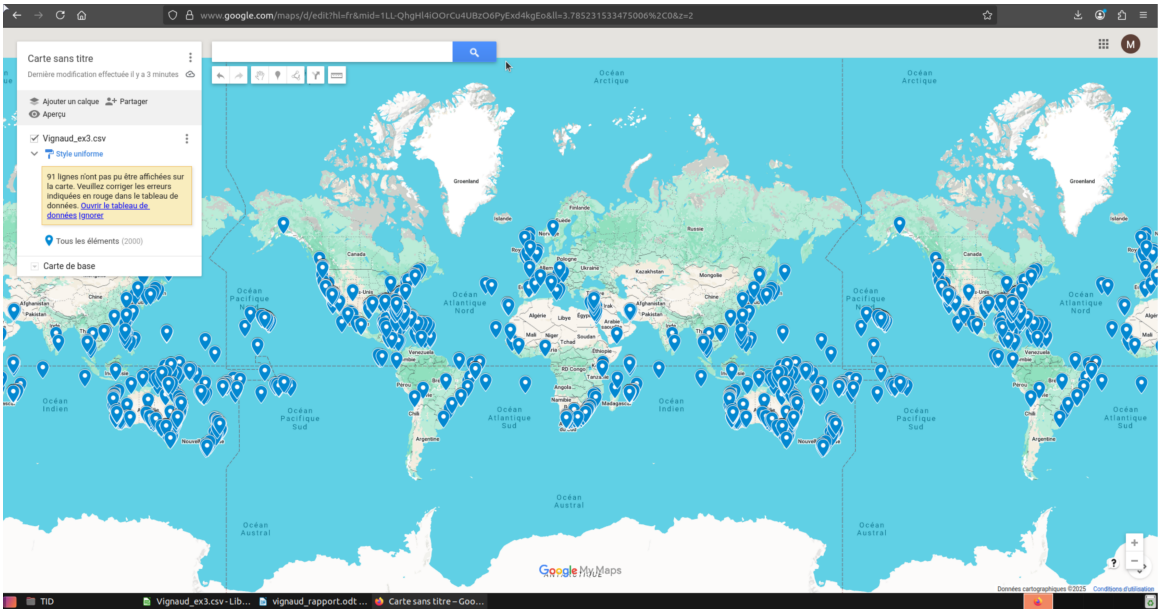


Figure 2: Worldwide distribution of attacks (map made with Google My Maps from a UTF-8 CSV export).

Rank	Country	Attacks	Most affected region	Attacks
1	United States	2,158	Florida	1,015
2	Australia	1,301	New South Wales	475
3	South Africa	571	KwaZulu-Natal	212

Table 1: Most affected countries and regions.

5 When and during which activity?

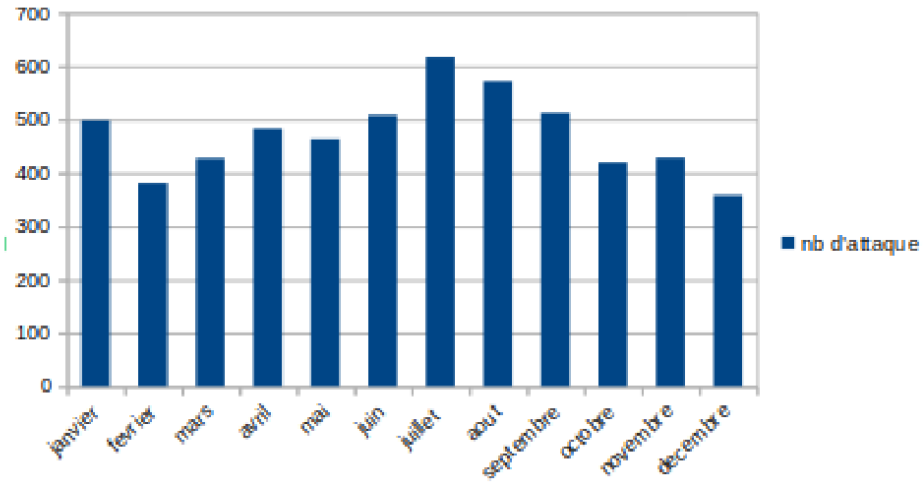


Figure 3: Number of attacks by month of the year (since 1900).

Months. Attacks are spread throughout the year, but with a summer peak in **July–August** (holidays, warm water, crowded beaches). Surfing, however, brings autumn forward: when crossed with activities, **October** appears as a high-risk month (177 attacks), a period of autumn swell when surfers are numerous.

Times of day. Attacks are concentrated in the late morning and the afternoon. In the United States in October, the riskiest time slot is **11:00–11:30 a.m.**

Activities. The top 3 is **surfing**, **swimming** and **fishing**. Fishing is the only one of these activities where attacks are mostly *provoked* (which makes sense), whereas for surfing and swimming they are almost always unprovoked. Fishing is affected all year round, swimming mostly from June to September, and surfing mostly in October–November.

6 Conclusion

So, is the shark a wolf to man? If we trust the results, yes: attacks are mostly unprovoked, clearly increasing since 1900, and highly localised. Indeed, if you go surfing on an October afternoon at New Smyrna Beach, Florida, you are more likely to end up as duck confit for a shark than to catch a good wave. But at the same time, no: after all, it is an animal, and we are the ones bothering it in its own home. The rise in attacks mainly follows the rise in beach attendance (tourism, water sports, and perhaps global warming bringing sharks closer to the coast). You don't really need to be unlucky to become a victim, just to be in the wrong place at the wrong time; and of course, in summer there are more people in the water, so more chances of getting chomped.